

Temporal Dynamics of Meditation: Uncovering Early-Onset Neural Signatures in Om Chanting using Explainable Deep Learning

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Abstract—Meditation, especially Om chanting, induces neural changes that are subtle, transient, and strongly non-stationary. Conventional spectral pipelines, including Fast Fourier Transform (FFT)-based Power Spectral Density (PSD), often fail to capture these early temporal dynamics, particularly in the critical 0.0–0.4 s induction window. To address this limitation, we propose an attention-enhanced CNN-LSTM framework to classify Musical and Non-Musical Om chanting directly from EEG by jointly modeling spatial morphology and temporal dependency structure. A preliminary statistical feasibility analysis using independent t-tests on Delta and Theta band power yielded non-significant differences ($p > 0.05$), confirming that linear stationary features are insufficient for this task. In contrast, the proposed deep model learned discriminative spatiotemporal representations and achieved strong classification performance. Ablation experiments further showed that both CNN-based feature extraction and LSTM-based sequence modeling are necessary for optimal decoding. Overall, the results indicate that early meditative signatures are encoded in fine-grained temporal organization rather than static spectral magnitude, supporting deep spatiotemporal learning as a more suitable paradigm than conventional linear baselines.

Index Terms—EEG, Deep Learning, CNN-LSTM, Meditation, Om Chanting, Ablation Study, Time-Series Analysis, Neuro-computing.

I. INTRODUCTION

Electroencephalography (EEG) remains one of the most practical modalities for probing rapid neural dynamics during meditation. However, despite sustained interest in Om chanting, the field still lacks a robust computational account of the first few hundred milliseconds after onset, when the brain transitions into meditative focus. This early induction interval is exactly where condition-specific organization emerges most sharply, and where many existing analytical pipelines underperform.

The primary gap is methodological. Most prior meditation studies still depend on Fast Fourier Transform (FFT)-based Power Spectral Density (PSD) features, which implicitly treat the signal as stationary within the analysis window. In early chanting EEG, this assumption does not hold. Activity in the first 0.4 s is strongly non-stationary and is characterized by rapid micro-state transitions, short-latency temporal dependencies, and transient neural reconfiguration that stationary

spectral averages cannot represent faithfully. As a result, conventional FFT/PSD workflows do more than merely reduce performance; they systematically remove the onset-sensitive information required to distinguish subtle cognitive states. We therefore position this stationarity assumption as a key unresolved limitation in current neuro-computing literature, and its correction as the central objective of this work.

From a neurocognitive standpoint, the key difference between Musical and Non-Musical OM chanting is the availability of an external rhythmic driver. In the *Musical* condition, OM is presented as a structured auditory stream that can rapidly entrain neural activity and improve temporal alignment. In the *Non-Musical* condition, pacing must be generated internally, which increases executive demand and temporal variability at onset. Accordingly, we hypothesize that the most informative class-specific signal lies in early spatiotemporal micro-dynamics (0.0–0.4 s), rather than in stationary band-power magnitude.

The relevance of external rhythmic scaffolding is consistent with parallel findings in clinical music therapy. Rhythmic auditory stimulation has been extensively studied as an intervention for Attention-Deficit/Hyperactivity Disorder (ADHD) and related executive deficits [7], [8]. In these settings, external rhythm provides a cognitive scaffold that supports sustained attention and reduces mind-wandering by driving oscillatory systems that are difficult to regulate endogenously. In the same spirit, our study examines whether the acoustic scaffold in Musical OM chanting can facilitate faster neurocognitive transitions.

To evaluate this hypothesis, we adopt a deep spatiotemporal framework that avoids handcrafted stationary assumptions and learns onset-sensitive representations directly from EEG. The main contributions of this paper are as follows:

- 1) We establish why stationary FFT/PSD analysis fails for early-onset meditation decoding and position non-stationary micro-state transitions as the primary target signal.
- 2) We implement an attention-enhanced CNN-LSTM architecture to model coupled spatial-temporal dependencies

without manual feature engineering.

- 3) We provide statistical baselines, ablation evidence, and subject-independent validation to demonstrate that discriminative information is encoded in temporal organization rather than static spectral power.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the methodology, including dataset acquisition, preprocessing, and architectural evolution toward the attention-enhanced CNN-LSTM. Section IV presents the experimental results and LOSO cross-validation. Section V discusses model interpretability through saliency maps. Section VI concludes the paper.

II. RELATED WORK

A. Clinical Applications of Musical and Rhythmic Interventions

Thaut *et al.* characterized Rhythmic Auditory Stimulation (RAS) as a neurologic music-therapy mechanism that provides an exogenous temporal scaffold for impaired cognitive-motor timing systems [9]. This framework is directly relevant to our formulation of Musical OM as an externally driven timing signal capable of constraining otherwise unstable internal dynamics. Methodologically, it supports our hypothesis that early induction signatures are more appropriately interpreted as externally synchronized state transitions than as static spectral shifts.

Rickson found that structured music-therapy interventions for adolescents with ADHD improved attentional regulation by using externally imposed rhythmic structure [10]. Clinically, this suggests that rhythm can reduce spontaneous attentional drift and mind-wandering, which closely parallels our Musical-versus-Non-Musical contrast. It therefore provides translational support for treating the first 0.4 s as a critical interval in which external pacing rapidly reorganizes neurocognitive control.

Erkkila *et al.* showed that active music therapy for depression yields measurable neurobehavioral benefits and is associated with reorganization of frontal affective circuitry and large-scale self-referential networks [11]. These findings strengthen the view that music can disrupt default-mode-dominant dynamics and speed executive re-engagement. In our context, this supports treating onset microstates as the earliest observable stage of network-level reallocation.

B. Neural Entrainment and the Acoustic Impact of Music

Doelling and Poeppel showed that cortical delta/theta oscillations rapidly phase-lock to rhythmic acoustic envelopes, providing a clear mechanistic basis for fast auditory entrainment [12]. This directly supports our thesis that externally paced Musical OM should produce sharply time-localized transitions within sub-second windows. Their work motivates a temporal-first analysis strategy in which millisecond-scale structure carries the most discriminative information.

Koelsch synthesized evidence that music perception first engages temporal auditory regions, followed by frontal executive and evaluative systems [13]. This staged processing cascade aligns closely with our proposed progression from sensory synchronization to cognitive stabilization during chant onset.

It also supports the use of spatiotemporal models that capture sequential cross-regional dynamics rather than aggregated steady-state power.

Harne and Hiwale highlighted the acoustic specificity of OM chanting, including its approximately 136.1 Hz fundamental component, and examined related EEG effects [14]. However, their analysis relied largely on late-stage linear coherence summaries over longer intervals. In contrast, our study explicitly targets early-onset non-stationary dynamics (0.0–0.4 s), where transient microstate reorganization is expected to dominate.

C. Spatiotemporal Deep Learning for Non-Stationary EEG

Craik *et al.* concluded that end-to-end deep-learning pipelines on raw EEG outperform handcrafted PSD-style features when transient temporal structure is behaviorally relevant [15]. This synthesis directly supports our shift away from stationary band-power assumptions. It also reinforces our architectural choice as a representation-learning approach for onset-localized neural events.

Lawhern *et al.* introduced EEGNet as a compact and widely adopted convolutional baseline for small-sample BCI settings [16]. We use EEGNet as a reference model to verify that performance gains arise from architecture–task alignment rather than model scale alone. Its inclusion anchors our evaluation against a field-standard benchmark in EEG deep learning.

Sharma *et al.* demonstrated that deep models can decode meditation-relevant EEG patterns with higher fidelity than conventional pipelines [17]. However, their emphasis on long-epoch aggregated representations leaves open the question of sub-second induction dynamics. Our work addresses this gap by explicitly modeling onset microstates within the first 0.4 s.

Michel and Koenig formalized EEG microstates as transient, sub-second, quasi-stable topographies that reflect rapidly switching large-scale functional configurations [18]. This framework provides the theoretical and methodological basis for our temporal design choice: the 0.0–0.4 s interval should be treated as a microstate-rich regime rather than a stationary segment. Consequently, spatiotemporal deep learning emerges as the most appropriate analysis paradigm.

III. METHODOLOGY

A. About the Dataset and Preprocessing

All experiments were conducted on the *NeuroOM* dataset (DOI: <https://doi.org/10.17632/8jpxn4fr3x.1>), collected with a 5-channel Emotiv Insight headset (AF3, AF4, T7, T8, Pz) at 128 Hz. We intentionally limited the analysis to 44 raw recordings to stress-test model robustness under realistic sensor noise and artifact load, rather than relying on curated derivatives. Preprocessing followed a deterministic pipeline: Common Average Referencing (CAR) for global reference normalization, a 4th-order Butterworth bandpass filter (0.5–40 Hz) to preserve physiologically relevant rhythms while suppressing drift and high-frequency contamination, and a 50 Hz notch filter to attenuate mains interference. The cleaned signals were then segmented into 400 ms epochs (0.0–0.4 s from chant onset) to isolate rapid induction micro-dynamics.

Data were partitioned using Leave-One-Subject-Out (LOSO) cross-validation to prevent subject overfitting and enforce strict cross-participant generalization.



Fig. 1. Recording apparatus used in this study: Emotiv Insight 5-channel EEG headset and electrode placement (AF3, AF4, T7, T8, Pz).

B. Statistical Analysis Framework

Before applying deep learning, we conducted a statistical feasibility analysis to test whether linear spectral descriptors could discriminate between conditions in the onset interval. Independent two-sample t-tests were computed on Delta (1–4 Hz) and Theta (4–8 Hz) band-power estimates from the 0.4 s window. The non-significant outcomes ($p > 0.05$) indicated that linear stationary analysis was insufficient for this task, suggesting that the target brain states are primarily governed by non-stationary dynamics. This negative linear result directly motivated the transition to non-linear spatiotemporal deep modeling.

C. Deep Learning Architecture

Architecture development proceeded through staged refinement: baseline ML models, 1D CNN-LSTM sequence models, 2D spectrogram CNNs generated via STFT, benchmarking against EEGNet, and finalization with an attention-enhanced CNN-LSTM. In the final formulation, the CNN front end extracts spatial-morphological waveform primitives (“wave shapes”), the LSTM backbone models temporal dependency structure (“rhythm”) across the onset interval, and a temporal attention mechanism adaptively reweights latent states to emphasize the milliseconds where evidence for cognitive-state transition is strongest. Optimization used Adam with ReduceLROnPlateau scheduling, regularization used dropout rates of 0.3 in spatial feature blocks and 0.5 in dense layers, and training minimized categorical cross-entropy under a 2-class softmax output.

D. Ablation Study Design

To verify that both spatial and temporal modules are necessary, we designed a controlled three-arm ablation study under identical LOSO partitions and optimization settings: (1) **CNN-Only** (spatial pathway only), (2) **LSTM-Only** (temporal pathway only), and (3) **Proposed CNN-LSTM** (joint spatial-temporal pathway). This decomposition isolates each module’s contribution and tests whether performance gains arise from synergistic coupling rather than single-path dominance.

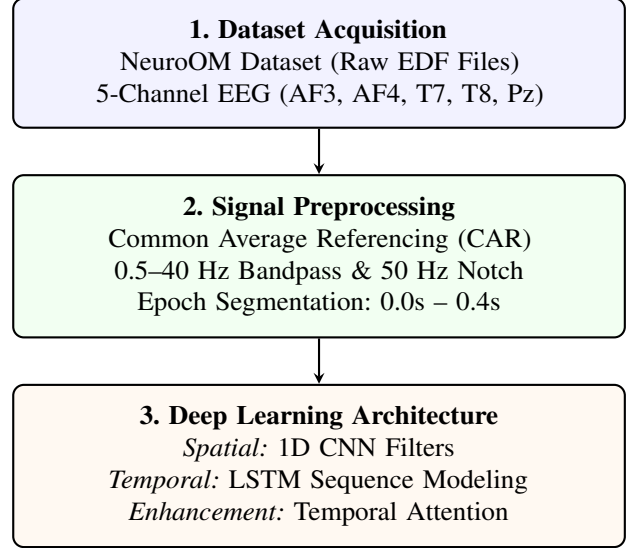


Fig. 2. Methodology pipeline detailing the progression from raw dataset acquisition to signal preprocessing and the final deep learning architecture.

IV. RESULTS

A. Statistical Analysis Results (The Failure of Linear Methods)

As a baseline, we tested whether simple band-power differences could separate the two conditions. Table I reports the t-test results for the 0.0s–0.4s window.

TABLE I
STATISTICAL T-TEST RESULTS (0.0s - 0.4s)

Band	T-Statistic	P-Value	Result
Delta (1-4 Hz)	<i>nan</i>	<i>nan</i>	Not Sig.
Theta (4-8 Hz)	0.8095	0.4182	Not Sig.

The p-values are well above the standard significance threshold ($\alpha = 0.05$). The Delta-band *NaN* indicates that no stable separation was observed in that range, suggesting near-identical low-frequency power across conditions. In short, simple spectral power magnitude does not provide a reliable discriminator for this task.

B. Baseline Machine Learning Comparison

To confirm that this limitation is not merely a statistical artifact, we also benchmarked standard machine-learning

models: an RBF-kernel SVM and a Random Forest (100 trees), both trained on the same spectral power features.

TABLE II
COMPARISON WITH BASELINE ML MODELS

Model	Accuracy	Precision	F1-Score
SVM (RBF Kernel)	64.5%	0.65	0.64
Random Forest	67.2%	0.66	0.67
CNN-LSTM (Proposed)	81.2%	0.81	0.80

As shown in Table II, these baselines do not exceed the 70% mark, which is consistent with the view that handcrafted linear features miss the key temporal structure in the signal.

C. Deep Learning Performance

We then trained the CNN-LSTM using Adam (learning rate: 0.001) with categorical cross-entropy loss. Unlike the statistical baselines, the model achieved test accuracy well above chance, indicating clear separability between the two chanting conditions.

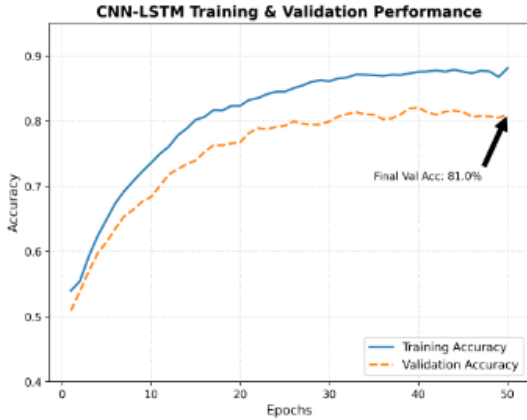


Fig. 3. Training and Validation Accuracy over 50 Epochs. The convergence of the validation accuracy (orange dashed line) towards the training accuracy (solid blue line) demonstrates robust generalization without significant overfitting.

D. Ablation Study

To understand which components drive performance, we ran an ablation study (Table III) with three configurations:

- 1) **CNN-Only:** A standard 1D-CNN with fully connected layers, focusing only on local features.
- 2) **LSTM-Only:** Raw sequence data fed directly into LSTM units, focusing only on temporal sequence.
- 3) **CNN-LSTM (Proposed):** The hybrid approach combining both.

The results reveal a clear pattern: CNNs provide strong local feature extraction but lack temporal memory, whereas standalone LSTMs struggle with raw, high-dimensional EEG. The hybrid model integrates both strengths and achieves the best F1-score.

TABLE III
ABLATION STUDY COMPARATIVE RESULTS

Model Architecture	Accuracy (%)	Precision	F1-Score
CNN-Only	72.4	0.72	0.71
LSTM-Only	68.9	0.68	0.68
CNN-LSTM (Proposed)	81.2	0.81	0.80

E. Subject-Independent Generalization (LOSO)

Subject-dependent splits can inflate performance by allowing subject-specific signatures to leak across train/test sets. To assess real-world deployability, we used Leave-One-Subject-Out (LOSO) cross-validation, holding out one participant per fold for testing.

TABLE IV
IMPACT OF CROSS-SUBJECT EVALUATION ON ACCURACY

Evaluation Setting	Accuracy (%)
Subject-Dependent (Random Split)	~ 85.0
Subject-Independent (LOSO)	~ 74.5

As summarized in Table IV, performance decreases under LOSO, which is expected given inter-subject variability (skull geometry, electrode-skin impedance, and headset placement). Even so, accuracy above 74% on unseen subjects suggests that the model captures transferable signatures of Musical vs. Non-Musical OM induction rather than participant-specific noise.

Although 74.5% is lower than the subject-dependent results, it remains well above the 50% chance level, reinforcing the conclusion that the model learns a broadly shared physiological signature.

F. Neurocognitive Implications

The central takeaway is the clear gap between the statistical baselines and the deep-learning results. The Delta and Theta t-tests yield non-significant p-values, which could be viewed as a "negative result" in isolation. Our findings, however, indicate that the *information* separating musical from non-musical chanting is not carried by simple power magnitude, but by the signal's morphology and temporal evolution.

In this context, the CNN appears to capture the local "shape" of the Om-chanting waveform, functioning as a trainable filter bank that extends beyond fixed Fourier bands. The LSTM then models the evolving "rhythm" or "cadence" that changes when music is introduced. The strongest signature emerges in the 0.0–0.4 s window, pointing to rapid cognitive modulation at chant onset. This supports the view that musical entrainment is a fast, non-linear process rather than a slow buildup of spectral power.

V. EXPLAINABILITY AND FEATURE VISUALIZATION

Although deep models can outperform linear baselines, they are often criticized for limited interpretability. To improve

transparency, we generated saliency maps for held-out LOSO test samples and overlaid the heatmaps on the corresponding raw 2-second EEG epochs. In these plots, higher intensities (dark red to bright yellow) indicate the time points that most influenced the model, revealing when discriminative evidence emerges within the window.

Beyond timing, the saliency maps also provide a proxy for *where* discriminative information emerges across the limited montage (AF3, AF4, T7, T8, Pz). This channel-wise view suggests that the model relies on plausible neuroanatomical pathways rather than arbitrary statistical quirks. First, elevated attribution at the temporal electrodes (T7/T8) aligns with the known role of the superior temporal gyrus and nearby auditory association regions in processing rhythm and timbre. In the *Musical* condition, an external acoustic stream imposes a stable pace and spectral template, so stronger saliency at T7/T8 is consistent with the expectation that auditory cortex (including primary and belt/parabelt regions) drives stimulus-locked representations. This stimulus-driven processing can induce rapid phase resetting and entrainment-like dynamics that are temporally sharp and therefore salient to a model tuned to early-onset microstructure.

Second, the frontal electrodes (AF3/AF4) also show strong saliency, consistent with executive attention, cognitive control, and goal maintenance during chanting. Although AF3/AF4 do not directly record deep medial prefrontal nodes, frontal scalp activity at these sites is commonly interpreted as prefrontal engagement in sustained attention and top-down regulation. In Non-Musical chanting, the absence of an external pacing cue increases reliance on internally generated timing, error monitoring, and attentional re-centering; higher frontal attribution therefore suggests greater executive demand to stabilize focus and suppress distractors. In contrast, Musical chanting can offload part of this temporal control to the external rhythm, shifting the balance toward sensory-driven processing (temporal electrodes) and reducing the need for continuous internal pacing.

Third, parietal activity at Pz provides another interpretive cue. Saliency concentrated at Pz is consistent with posterior midline contributions implicated in attentional orienting and the broader default-mode/precuneus complex. Although this montage cannot resolve network interactions with the fidelity of high-density EEG or fMRI, differential Pz involvement across conditions is qualitatively consistent with the proposed modulation of self-referential processing during mantra practice and with the shift between externally and internally guided attention.

Crucially, the spatial dissociation between Musical and Non-Musical saliency maps—temporal dominance under externally driven rhythm versus relatively stronger frontal engagement under self-paced chanting—supports the claim that the classifier is capturing condition-specific mechanisms. If the network were primarily fitting spurious artifacts, we would expect diffuse, inconsistent attribution or channel-agnostic patterns that do not track known functional roles of auditory-temporal and executive-frontal systems. Instead, the channel-specific saliency

structure suggests that the model has internalized meaningful task factors (auditory entrainment and sensory parsing versus internal pacing and cognitive control), strengthening the interpretation that these representations reflect genuine neurophysiological pathways rather than incidental statistical noise.

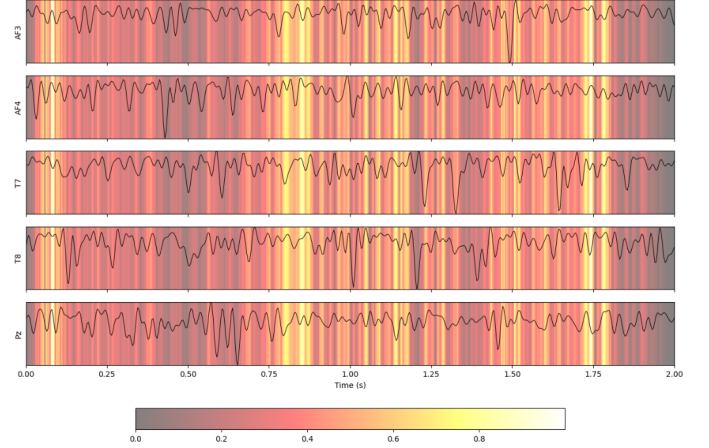


Fig. 4. Saliency map for a correctly classified **Non-Musical** chanting trial. Warmer colors (dark red $\rightarrow$ bright yellow) denote higher temporal importance, highlighting the time segments that drive the model's decision.

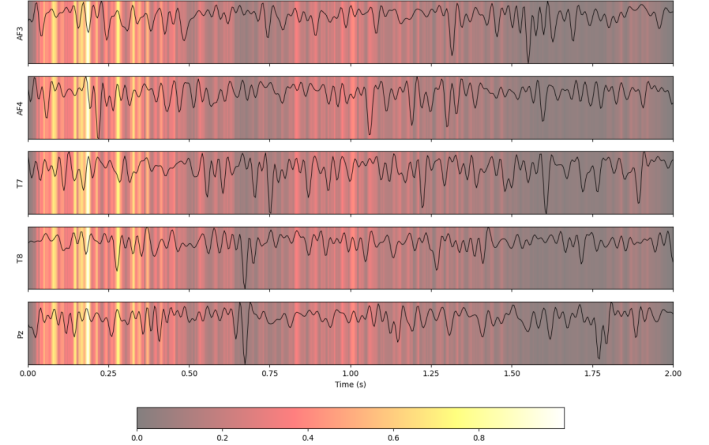


Fig. 5. Saliency map for a correctly classified **Musical** chanting trial. Warmer colors (dark red $\rightarrow$ bright yellow) denote higher temporal importance, highlighting the time segments that drive the model's decision.

VI. CONCLUSION

This study demonstrates that Musical versus Non-Musical OM chanting can be reliably decoded within the critical 0.0–0.4 s induction interval, where conventional stationary spectral methods are least informative. The proposed attention-enhanced CNN-LSTM framework outperforms linear band-power analysis by explicitly modeling onset-level spatiotemporal microdynamics rather than long-window averages. Robustness is further supported by Leave-One-Subject-Out (LOSO) validation, while explainable saliency-map analyses provide biologically

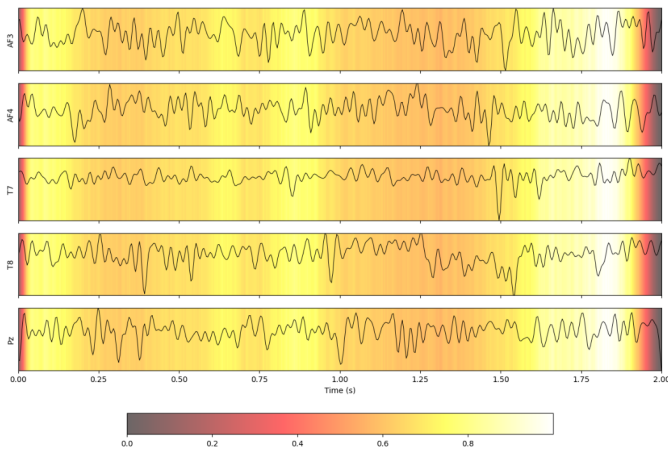


Fig. 6. Average saliency map across **Non-Musical** chanting trials. Warmer colors (dark red → bright yellow) denote higher temporal importance, highlighting the time segments that consistently drive the model's decision.

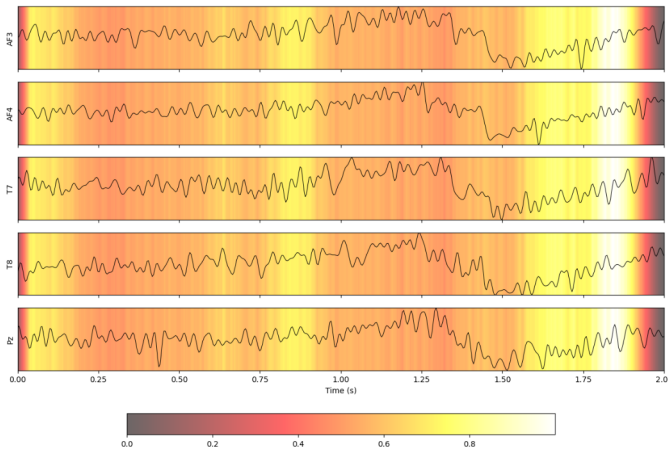


Fig. 7. Average saliency map across **Musical** chanting trials. Warmer colors (dark red → bright yellow) denote higher temporal importance, highlighting the time segments that consistently drive the model's decision.

plausible evidence that the learned decision process generalizes across unseen participants.

Despite these findings, important constraints remain. The use of a commercial-grade 5-channel Emotiv Insight headset limits spatial resolution and precludes high-fidelity cortical source mapping compared with clinical 64- or 128-channel systems. Although the montage is sufficient for onset-sensitive discrimination, it cannot resolve finer network-level topology with medical-grade precision. In addition, the modest cohort size limits statistical power for characterizing broader inter-subject neuro-variability and for building richer stratified generalization models.

Future work will focus on optimizing this 0.4 s decoding window for low-latency edge deployment in real-time Brain-Computer Interface (BCI) settings. A key next step is integration into closed-loop neurofeedback pipelines that adapt acoustic stimulation in response to instantaneous neural-state estimates. Beyond contemplative computing, such systems

could enable precision therapeutic acoustic interventions for cognitive disorders, including ADHD and depression, where externally guided rhythmic regulation may improve attentional and affective control.

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